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Monkspaces Compliance Platform – Architecture & Phased Implementation

The proposed SaaS platform will centralize product and packaging data from all sources (Trace One PLM, Snowflake data lake, spreadsheets, supplier inputs) into a **shared data foundation**, then apply data-quality checks and modular compliance logic (EPR, PPWR, DPP, etc.). This ensures that once data is cleaned and validated in one place, multiple compliance modules can reuse it without re-importing. The outline below explains the architecture, data flows, and build plan in detail, with diagrams and references.

1. Shared Data Foundation (Ingestion & Standardization)

- **Sources:** Product/packaging data will come from existing systems: e.g. Trace One PLM (Devex), a Snowflake data warehouse, or CSV/Excel uploads from suppliers. We'll use connectors or batch imports to ingest these into Monkspaces.
- **Normalization:** Imported data is mapped into a common schema (products, packaging components, materials, suppliers, sales volumes, etc.). For example, Trace One's REST APIs are designed to "streamline operations" and connect product data across ERP, PLM and other systems【9†L139-L142】. Similarly, Snowflake offers services like **Snowpipe** for continuous data ingestion and the new **Snowflake Openflow** data integration service (built on Apache NiFi) to move data between enterprise apps and the Snowflake "Data Cloud"【36†L235-L237】【36†L243-L249】. We will configure these or write import jobs to keep Monkspaces in sync with source data.
- **Data Model:** Key entities include *Product*, *PackagingComponent* (multiple per product), *Material*, *Jurisdiction*, *SalesRecord* (volume by market), *Supplier*, etc. For each packaging component we store material type(s), weight, unit, and associated product. The Monkspaces database also tracks *RuleSet* versions for each jurisdiction and *CalculationRun* results. (For example, an EPR calculation run references a specific rule version and the input weights/quantities.) This one schema underpins all modules.
- **Diagram:** *Business Workflow (Data to Insights)* – the flowchart below illustrates the end-to-end process: importing data, validating it, correcting issues, then running jurisdiction-specific EPR calculations and reporting results.

【36†L235-L237】【23†L626-L634】 *Figure: End-to-end data and compliance workflow. Source data is ingested and normalized, then data-quality checks produce issues (if any). After corrections, valid data feeds the selected jurisdiction's EPR calculation rules, yielding fee breakdowns, hotspots and scenarios.*

2. Data Quality & Correction Engine

- **Validation Rules:** Every record is checked for completeness and consistency. For example, mandatory fields (material, weight, unit, market) must be present; numeric fields must be positive; sum of component weights should match total packaging weight; materials must map to allowed categories. Special compliance checks (e.g. recyclability attributes) are also applied. If a supplier upload omits a required field, the system will flag it rather than assume a default.
- **Issue Tracking:** Any validation error creates a *Data Issue* record. Issues are triaged by severity and assigned to the responsible owner (e.g. internal analyst or the supplier). The workflow then

sends a correction request for missing or invalid values. This mirrors industry best practice: for instance, Certivo's EPR platform "parses declarations to extract material types, weights, and recyclability attributes…validates against PPWR material categories…and flags data gaps automatically"【23†L626-L634】.

- **Correction Loop:** When the owner submits corrected data (often with supporting evidence like a supplier certificate), an approver reviews and approves it. Upon approval, the system updates the record (with versioning) and reruns validation. Approved fixes clear the issue. A full audit trail is kept (who changed what, when). *Importantly*, writing corrections back to the original source (e.g. pushing updates into Trace One) requires additional integration support. We will plan write-back as a future enhancement only if the source system's APIs/permissions allow it.
- **Outputs:** Records that pass validation feed the compliance modules. Any unresolved issues remain in the queue for manual attention.

3. Compliance Modules (EPR, PPWR, DPP, etc.)

- **EPR Fee Calculation:** This is the first vertical slice. The user selects a **jurisdiction** (e.g. "Germany 2025") and reporting period. The platform loads that rule set version and applies it to the validated product data: it groups packaging quantities by material/category, applies the scheme's rates and eco-modulation factors, and sums up fees. Intermediate results (component-level fees, totals by product and material) are stored. Users see an explainable breakdown of how each fee was computed. This matches the conceptual formula for EPR: sum of (eligible quantity × tariff × adjustments). All inputs (selected rule version, input weights/volumes, mappings) are recorded so any result can be audited and reproduced.
- **Reuse of Common Data:** Once the shared data foundation is built, additional modules can reuse the same data. For example:
- **PPWR Compliance:** Uses product/packaging composition to calculate compliance measures (e.g. recyclability scores, mass balances) per EU Packaging and Packaging Waste Regulation requirements.
- **Digital Product Passport (DPP):** Uses product identity and material data to populate standard passport fields (materials, recycling info, etc.) for sharing with customers/authorities.
- **Packaging Costing:** Uses the same packaging structure to calculate costs (e.g. forecasting packaging material costs using linked financial data).
- **Other Modules:** Future modules (battery compliance, component traceability, etc.) can be added on the same foundation. Each module has its own logic but *no separate data import* is needed.
- **Dashboards & Reports:** The platform will offer dashboards summarizing the results: total EPR liability by jurisdiction, high-cost "hotspots" (products or materials contributing most to fees), and scenario tools. Users can alter assumptions (e.g. reduce plastic weight by 10%) and immediately see the impact on fees, following the "what-if analysis" practice. Finally, the system can generate formatted reports and submission packages. As one EPR solution notes, automated compliance tools can produce "jurisdiction-specific placed-on-market reports" and "audit-ready documentation" in hours rather than weeks【23†L655-L663】. Our MVP will aim to deliver similar outputs (e.g. Excel or XML submissions per national PRO requirements) with full traceability.

4. MVP Scope and Key Requirements

- **Minimum Viable Product (MVP):** Focus on one complete end-to-end workflow in a single jurisdiction. This includes: ingesting a representative product dataset; identifying and correcting data gaps; applying an approved EPR rule set for one packaging stream (e.g. plastic bottles in 2025 FR); and showing the fee breakdown, hotspot chart, and one basic scenario.

- **Core Features:** Data import/mapping; central database; data-quality engine; correction workflow; rule-version management; one calculation engine; results dashboard.
- **Exclusions (initially):** We will not attempt to implement all jurisdictions or all compliance types at once. Other compliance calculations (PPWR, DPP), advanced forecasting, or automated trace-out to every legacy system will come later. The priority is proving the core process works with real data.
- **Data Requirements:** We must gather sample data from KHC for products and packaging (component weights, materials, volumes by market). We also need at least one confirmed rule set (e.g. a French EPR tariff schedule) and expected output for validation. Without a real dataset and reference calculation, we can only demonstrate the workflow, not the final accuracy of fees.

5. Integration Considerations

- **Trace One (Devex) Integration:** We assume KHC's packaging data may reside in Trace One PLM after its migration. Trace One provides **REST APIs** for interoperability (connecting ERP, PLM, etc.) 【9†L139-L142】. We will evaluate its API documentation to extract product and packaging info. Possible approaches: periodic data export from Trace One into Monkspaces, or a live API sync. Until the scope is confirmed, this integration remains **unverified**. We plan it as optional: the MVP can start with file uploads or Snowflake if needed, and add the Trace One connector once access is arranged.
- **Snowflake Data Warehouse:** If Snowflake is used by KHC as a central data store or analytics warehouse, we can either query it directly or ingest from it. Snowflake supports many integration methods: it has **Snowpipe** for streaming data loads and now **Openflow** for broad ETL flows【36†L235-L237】【36†L243-L249】. We'll confirm what Snowflake schema/permissions KHC has. In any case, Snowflake will likely be used as an *analytics sink* (e.g. for running large-scale queries or staging data). Our initial design assumes a separate PostgreSQL for transactional data, with Snowflake serving read-only analytics if needed.
- **User Inputs:** For any data the user or supplier provides (Excel files, manual entry), the UI will map those fields into the system. We will design the import interface to be flexible (e.g. allow CSV column matching) so that even if automated connectors aren't ready, the MVP can still be fed sample data.
- **Write-Back to Sources:** Our initial MVP will treat Monkspaces as an append-only data store with respect to sources. If corrections are made, we will log them in the system but not push them automatically to Trace One or Snowflake (unless those APIs explicitly support updates and we have stakeholder approval).

6. Phased Implementation Roadmap

We recommend a staged build with clear milestones and decision gates. Each phase ends with a review before proceeding.

1. **Phase 0 – Discovery & Scope (2–4 weeks):** Confirm target use-case. Which jurisdiction (or country), what product categories, and what source system will be the "system of record" for data? Secure sample data extracts (Trace One export, Snowflake table, or spreadsheets). Identify a stakeholder who owns regulatory rules and obtain one set of EPR tariffs. *Exit Criteria:* Sample data loaded, integration approach agreed (file vs API), initial success in connecting to one data source.
2. **Phase 1 – Data Foundation (3–6 weeks):** Build the core database and import mechanism. Develop data ingestion pipelines (e.g. CSV loader, simple API connector) and create the canonical data model (product, packaging, materials, etc.). Implement tenant isolation (each customer's data segregated). Populate the database with test records. *Exit Criteria:* Data can be imported/

mapped into Monkspaces and viewed (e.g. via a “Product” table). Identifiers from source (Trace One ID, Snowflake table name) are preserved for lineage.

3. **Phase 2 – Data Quality & Issue Management (4–8 weeks):** Implement validation rules (required fields, ranges, consistency checks). Create the “Issue Queue” UI (filters by severity, owner). Code the correction workflow (assignment, evidence upload, approval). Integrate an audit log capturing changes. *Exit Criteria:* A sample missing-field case can be raised as an issue, corrected by a user, and then cleared by the system. The data-quality status of each record is tracked.
4. **Phase 3 – EPR Calculation Engine (4–6 weeks):** Integrate a rule-version module and coding of calculations. Load one jurisdiction’s rule set (from Phase 0), including material categories and tariffs. Build the calculation service that, for each valid record, computes fees. Display the fee breakdown by product/material on the UI. *Exit Criteria:* For the sample data, the system produces an EPR fee output that can be reconciled against a reference calculation for that jurisdiction. All inputs (rule version, volumes) are recorded.
5. **Phase 4 – Reporting & Analytics (3–5 weeks):** Develop dashboards for business insight. Implement cost hotspot charts (rank products or materials by fee), and a scenario builder (let user tweak an assumption and re-run the calculation). Enable exporting of results and gap reports. *Exit Criteria:* Stakeholders can view a dashboard summarizing total fees and can generate a report or data file for submission. Scenario changes produce visible results.
6. **Phase 5 – Pilot Hardening (3–4 weeks):** Stabilize the application for a pilot. Add security (role-based access, single sign-on if needed), error handling, unit/regression tests, and logging. Finalize user documentation. Demonstrate a pilot to KHC with their data. *Exit Criteria:* The platform has passed basic security reviews, and pilot users can log in, correct data, and run EPR scenarios successfully. A go/no-go decision is made.
7. **Phase 6 – Expansion (ongoing):** After a successful pilot, iteratively add more features/modules: additional jurisdictions, PPWR module, DPP reports, supplier portal for self-service data entry, etc. Continue building connectors (real-time sync with Trace One, automated Snowflake ingestion) and addressing any gaps.

7. Key Assumptions & Next Steps

- **Source System Confirmation:** We assume after the Devex Trace One migration, the target packaging data will indeed reside in Trace One (or be accessible via it). This must be confirmed (which version of Devex, what data fields it holds). Snowflake’s role must also be clarified (raw data lake vs. analytics database).
- **Jurisdiction & Data Sample:** We need KHC to pick an initial market/use-case and provide representative data. For example, if focusing on Germany’s EPR scheme, we need the German packaging regulations and sample sales data. The sample should include various packaging types to test the system.
- **Regulatory Rules Maintenance:** Jurisdiction-specific logic must be maintained in the system. We assume traceability for rule versions (who approves them, effective dates, etc.). Any change in law (new rates or categories) will be managed by updating or adding rule sets in the system.
- **Integration Limits:** We have not yet verified whether Trace One allows writing corrected data back, or if Snowflake can be used in a write-enabled manner. Until these are confirmed, Monkspaces will act as the master data store for compliance purposes, with manual sync to legacy systems if needed.
- **Incremental Value:** By building a modular SaaS platform, we avoid reinventing the wheel for each compliance report. Infosys can deliver the shared data hub and EPR calculator faster than an ad-hoc in-house solution, and the same architecture can scale to future requirements (different jurisdictions, new product categories, etc.).

The above plan provides a **clear blueprint**: a central data platform with built-in validation, plus plug-in compliance modules. Starting with one complete EPR workflow (data to fee report) demonstrates value immediately, while the phased approach and gating minimize risk. Subsequent modules (PPWR, DPP, cost-forecasting) will reuse the same data and engine. At each phase we will validate assumptions (data availability, rule correctness, system performance) before moving on.

Sources: Trace One PLM offers REST APIs for real-time data sync【9†L139-L142】. Snowflake supports continuous ingestion (Snowpipe) and a new NiFi-based integration service【36†L235-L237】【36†L243-L249】. Industry solutions like Certivo's CORA emphasize automated packaging-data extraction, validation and audit-ready reporting【23†L626-L634】【23†L655-L663】 – approaches which our design follows to ensure accuracy and traceability.